

Introduction:

Biomedical signal processing plays a vital role in modern healthcare systems by enabling the analysis of physiological signals such as electrocardiogram (ECG) and electroencephalogram (EEG). These signals provide critical information about cardiac and neural activity and are widely used in diagnostic, monitoring, and clinical decision-making systems. However, ECG and EEG signals are characterized by very low amplitudes and low frequencies, often in the millivolt or microvolt range, making accurate signal processing challenging.

One important operation in biomedical signal analysis is peak value detection, which is used in applications such as heart rate estimation, EEG feature extraction, and threshold-based diagnosis. Conventional peak detector circuits based on diodes suffer from significant inaccuracies when processing low-voltage signals due to diode forward voltage drop, leading to incorrect peak detection and loss of vital information.

To address these limitations, precision peak detection circuits using operational amplifiers are employed. By incorporating the diode within the feedback loop of an op-amp, the effect of diode voltage drop is minimized, enabling accurate peak detection even for low-amplitude signals. This project focuses on the design and implementation of a high-accuracy precision peak detection circuit using the LM358 operational amplifier. The circuit is designed to operate on a single supply and is suitable for low-frequency biomedical signals. The performance of the proposed circuit is analyzed through LTspice simulation and validated using hardware implementation.

Problem Statement:

Biomedical signals such as electrocardiogram (ECG) and electroencephalogram (EEG) are characterized by very low amplitudes, typically in the millivolt or microvolt range. Accurate extraction of peak information from these signals is essential for reliable diagnosis and monitoring in biomedical applications. However, due to their low signal levels and susceptibility to noise, precise signal processing becomes a challenging task.

Conventional peak detection circuits primarily use diode-based rectification techniques, which introduce significant errors when processing low-voltage signals. The forward voltage drop of the diode causes inaccurate peak detection, especially in applications involving weak biomedical signals. As a result, the detected peak values do not accurately represent the true signal characteristics, leading to reduced reliability in peak-based measurements.

Therefore, there is a need to design a high-accuracy precision peak detection circuit that can reliably extract peak information from low-voltage biomedical signals. The circuit should overcome the limitations of conventional peak detectors while maintaining simplicity, low power consumption, and cost-effectiveness, making it suitable for practical biomedical signal processing applications.

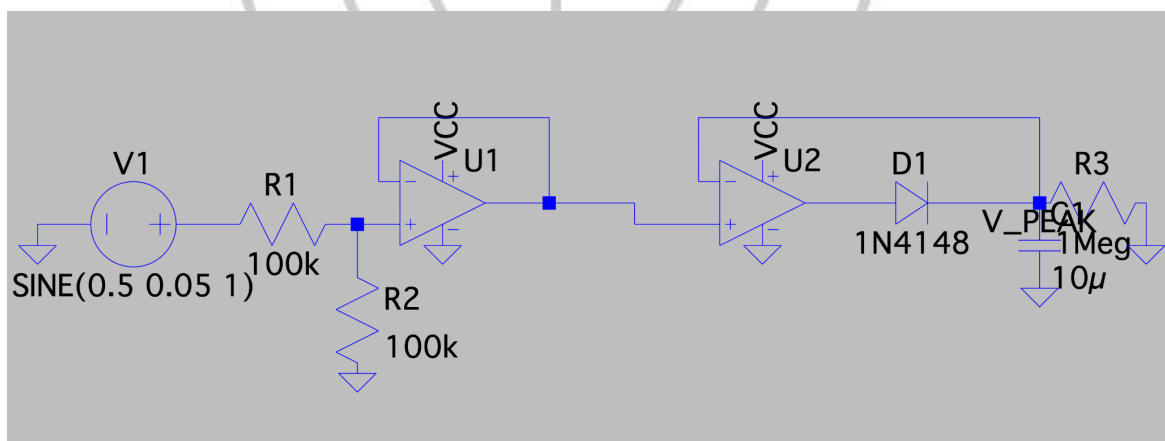


Fig 1: Schematic diagram of circuit on LTspice

Objectives:

The primary objectives of this project are:

- To design a precision peak detection circuit capable of extracting peak values from low-voltage biomedical signals such as ECG and EEG.
- To overcome the limitations of conventional diode-based peak detectors by using an op-amp-based precision rectifier.
- To implement a single-supply, low-power analog circuit suitable for biomedical applications.
- To simulate and verify the operation of the proposed circuit using LTspice for low-frequency, low-amplitude signals.
- To analyze the peak-hold and discharge behavior using an RC network.
- To demonstrate the suitability of the designed circuit for biomedical signal processing applications.

Methodology:

The design methodology followed a structured approach from signal analysis to simulation and hardware verification:

- **Signal Analysis:** The characteristics of low-voltage biomedical signals such as ECG and EEG were studied to understand their amplitude and frequency ranges. This analysis helped identify the challenges associated with extracting peak information from millivolt-level signals.
- **Conventional Peak Detector Study:** Traditional diode-based peak detector circuits were analyzed to understand their limitations. The effect of diode forward voltage drop on low-amplitude signals was identified as a major source of peak detection error.
- **Circuit Design:** A precision peak detection circuit was designed using the LM358 operational amplifier. The first stage was configured as a non-inverting voltage follower to buffer the input signal and prevent loading effects.
- **Precision Rectification:** An op-amp-based precision rectifier was implemented by placing the diode inside the feedback loop. This configuration minimizes the diode voltage drop and enables accurate peak tracking for low-voltage signals.
- **Peak-Hold Network Design:** A resistor–capacitor (RC) network was used to form the peak-hold stage. The capacitor stores the detected peak value, while the resistor controls the discharge rate of the stored voltage.
- **Simulation and Verification:** The complete circuit was simulated in LTspice using low-frequency, low-amplitude input signals. Input and output waveforms were analyzed to verify peak detection and hold behavior.
- **Hardware Implementation:** The designed circuit was implemented using discrete components and tested using a function generator and cathode ray oscilloscope (CRO). The hardware results were compared with simulation results to validate circuit performance.

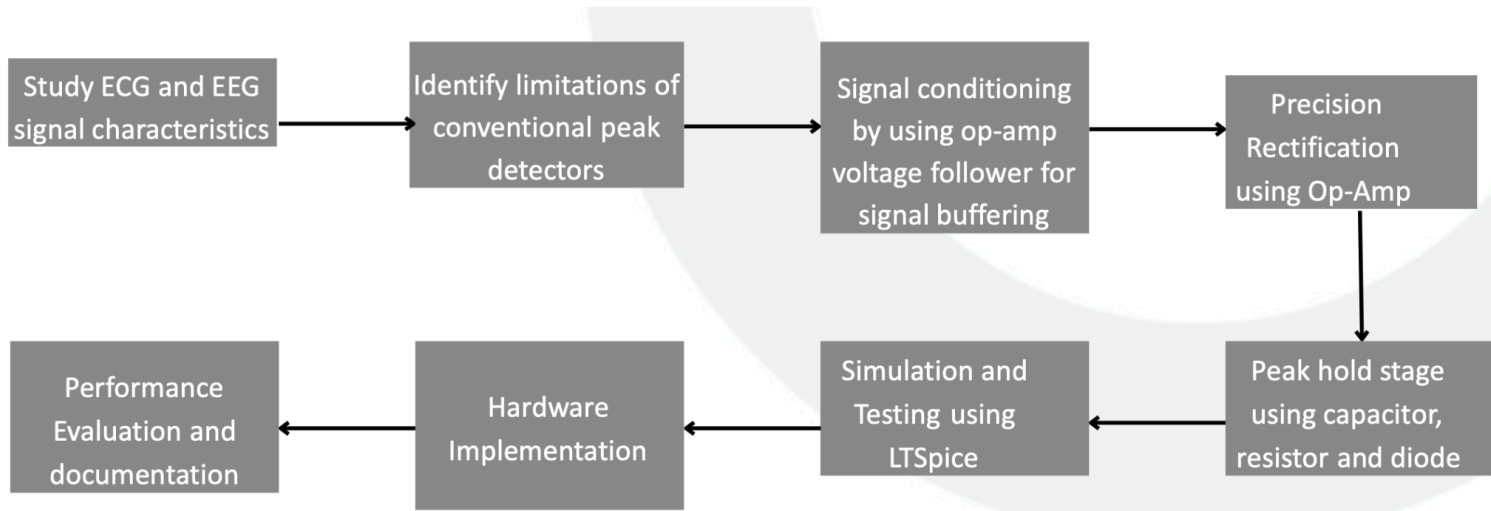


Fig 2: Methodology Flowchart

Hardware / Software tools used:

Hardware Tools:

1. **LM358 Operational Amplifier:** Used as the core active component for buffering and precision peak detection due to its single-supply operation and suitability for low-frequency signals.
2. **Diode (1N4148):** Used in the precision rectifier stage to enable accurate peak detection when placed inside the op-amp feedback loop.
3. **Passive Components:** Resistors and capacitors were used to form the input biasing network and RC peak-hold circuit for storing and discharging the detected peak value.
4. **DC Power Supply (+5 V):** Provided single-supply power to the LM358 operational amplifier.
5. **Function Generator:** Used to generate low-frequency, low-amplitude input signals representative of biomedical waveforms.
6. **Cathode Ray Oscilloscope (CRO):** Used to observe the input waveform and the peak-detected output signal.

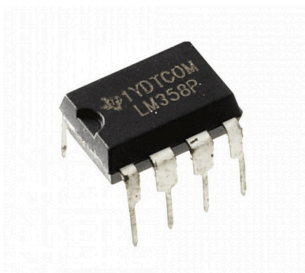


Fig 3: LM358 IC

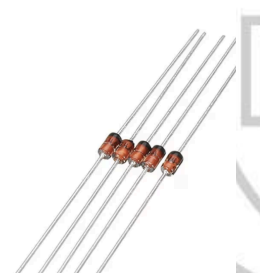


Fig 4: IN414



Fig 5: Signal Generator



Fig 6: Cathode Ray Oscilloscope

Software Tools:

- **LTspice:** Used for circuit schematic creation, transient simulation, and waveform analysis of the precision peak detection circuit.

- **KiCad:**
Used for schematic capture and PCB design of the proposed circuit.



Fig 7: LTspice Logo



Fig 8: KiCad Logo

Results:

Functional verification of the precision peak detection circuit was carried out through transient waveform simulation using LTspice, followed by hardware testing using a function generator and cathode ray oscilloscope (CRO). The simulation results, illustrated in the waveform plots, demonstrate the following observations:

- **Signal Conditioning and Buffering:** The voltage follower stage successfully buffered the low-amplitude input signal without distortion. The output of the buffer accurately tracked the input waveform, confirming high input impedance and effective isolation of the signal source.
- **Precision Rectification:** The op-amp based precision rectifier stage effectively eliminated the influence of diode forward voltage drop. Even for low-amplitude input signals, accurate rectification was achieved, validating the correctness of the feedback-based diode configuration.
- **Peak Detection and Hold Operation:** The peak-hold stage successfully captured the maximum value of the input signal. The capacitor charged to the peak voltage during each cycle, while the discharge resistor controlled the decay of the stored peak value, resulting in a slowly varying DC output corresponding to the signal peak.
- **Simulation Verification:** LTspice simulation results clearly showed the peak-detected output following the maximum value of the input waveform. The observed peak-hold behavior matched the theoretical operation of the circuit, confirming the correctness of the design.
- **Hardware Validation:** The circuit was implemented using discrete components and tested using a function generator and CRO. The hardware output demonstrated peak-hold behavior consistent with simulation results. Minor variations in waveform appearance were attributed to CRO scaling, component tolerances, and practical measurement limitations.

Overall, the results confirm that the designed precision peak detection circuit accurately extracts peak information from low-voltage, low-frequency signals, demonstrating its suitability for biomedical signal processing applications.

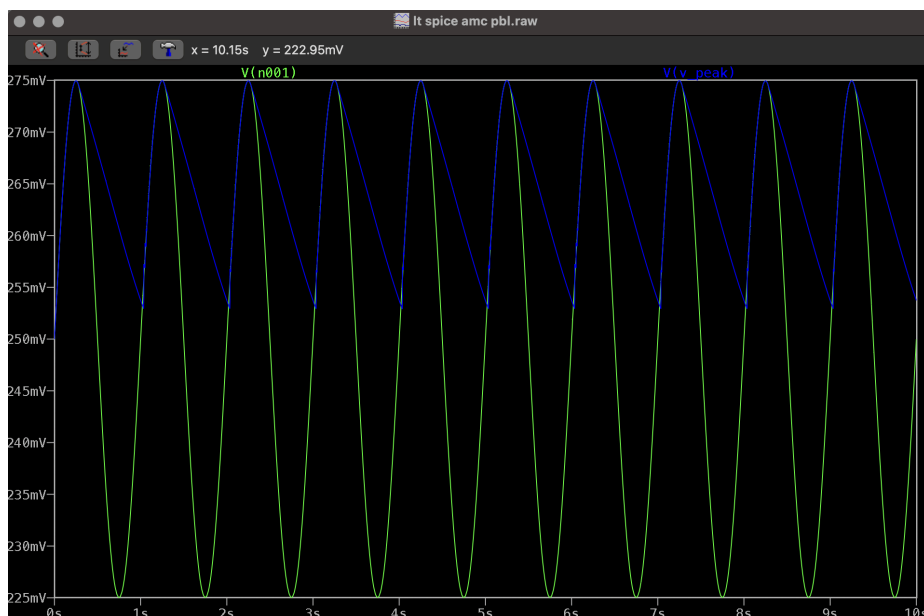


Fig 9: Graph of Input (green) and output (blue) where we can see the detected peaks

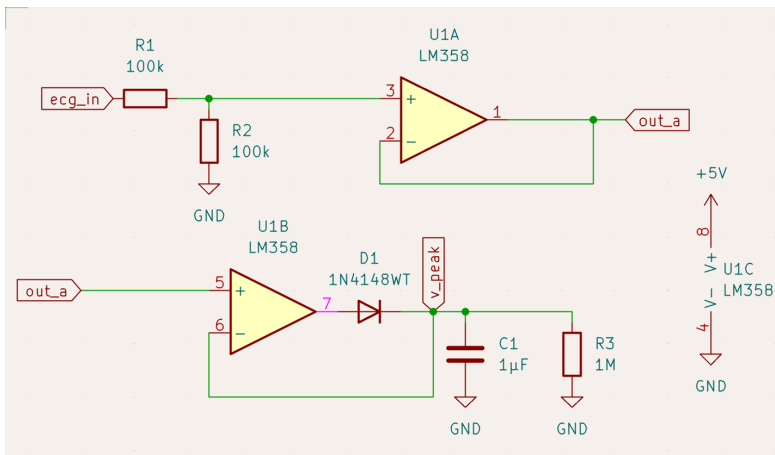


Fig 10: Schematic diagram on KiCad

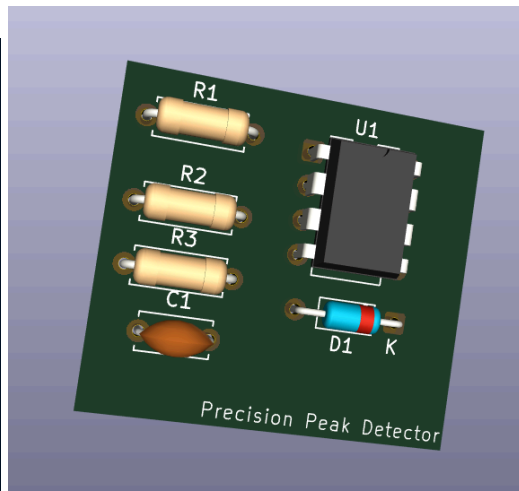
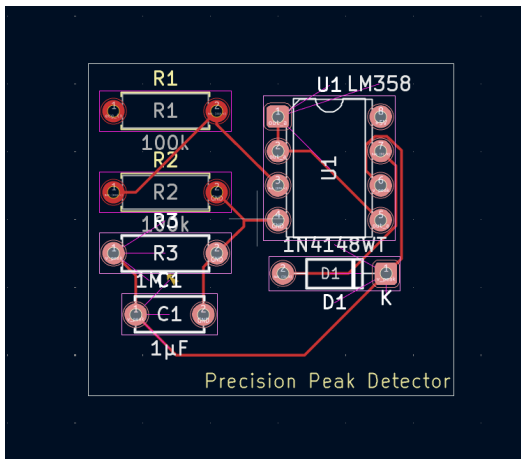


Fig 11&12: PCB Design

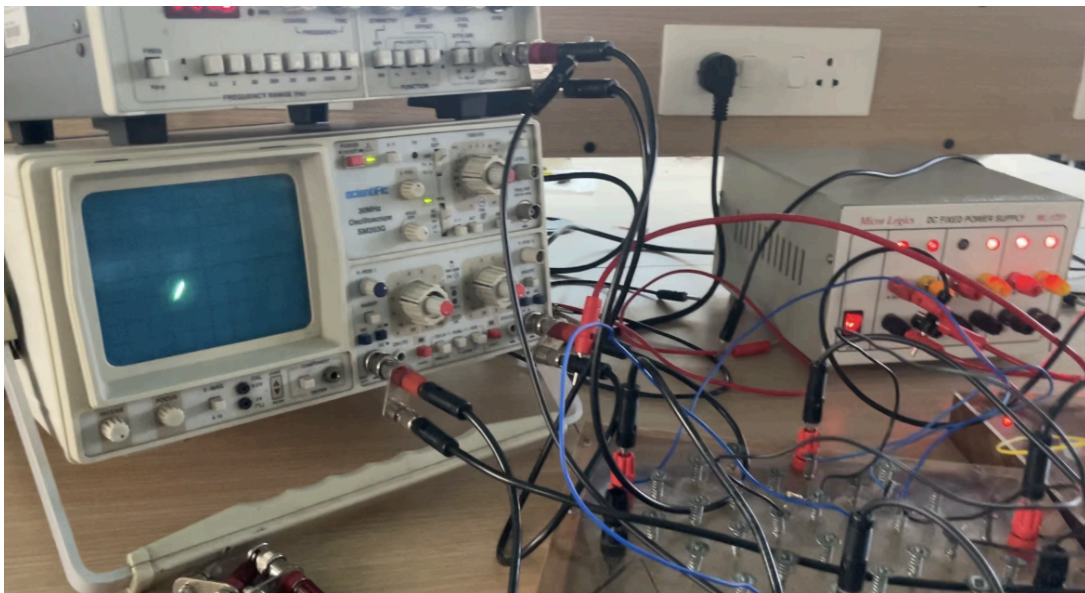


Fig 13: Hardware setup

Conclusion:

In this project, a high-accuracy precision peak detection circuit suitable for low-voltage biomedical signals was successfully designed and implemented. The circuit addressed the limitations of conventional diode-based peak detectors by employing an op-amp-based precision rectifier, thereby minimizing errors caused by diode forward voltage drop.

The use of the LM358 operational amplifier enabled single-supply, low-power operation, making the design appropriate for biomedical signal processing applications. Simulation results obtained using LTspice verified the correct operation of the buffer stage, precision rectification, and peak-hold functionality. The peak-hold RC network effectively captured and retained the maximum value of the input signal while providing controlled discharge behavior.

Hardware implementation further validated the functionality of the circuit, with observed results showing good agreement with simulation trends. Minor variations between simulation and hardware outputs were attributed to practical factors such as component tolerances and measurement limitations. Overall, the project demonstrates that the proposed precision peak detection circuit can reliably extract peak information from low-amplitude biomedical signals, confirming its suitability for applications involving ECG and EEG signal analysis.